

■# "PAPER_{0:D3}" -f [int]# PAPER #105 — BH Phases, Nebulae, and Emerging Astrophysical Domains

Author: Daniel T. Murphy

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Title: Black Hole Phases, Planetary Nebulae, and 10 Galaxy/Nebula UQFF Models: Complete §1.13 Validation Suite

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic **Date:** March 7, 2026 **Source Data:** validateddrawingsmodels.py (BHPHASESMODEL, Drawings 5–9); validateallmodels.py (10 galaxy models) **Index Slot:** §1.13 Multi-Physics Models, \$n = [int]# PAPER #105 — BH Phases, Nebulae, and Emerging Astrophysical Domains

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$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi G M c}{\kappa_{\text{es}}} \text{Bigl}(1 - [SSq] \cdot e^{-\kappa \Delta t} \text{Bigr}), \quad \text{quad } [SSq] = 0.57$$

<!-- ? = 5.0e-4 day¹, [SSq] = 0.57, β_i = 6.1e-1 -->

Abstract

This paper consolidates the remaining §1.13 validation results: (1) Black Hole Phase transitions (BHPHASESMODEL, Drawings 5–9; 5 phases), and (2) 10 galaxy/nebula UQFF models from validate_all_models.py (NGC2264, UGC10214, NGC4676, RedSpiderNebula, NGC3372, AGCarinae, M42, TarantulaNebula, NGC2841, MysticMountain). All models are implemented as calculator classes receiving system parameters; no hardcoded data. Combined, these represent the broadest validation sweep of UQFF across astrophysical environments.

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Part A: Black Hole Phase Model (BHPHASESMODEL, Drawings 5–9)

A.1 The 5 BH Phases

Drawing 5: stellar mass BH formation (collapse, Drawings 6–9 show phase transitions):

Phase	Drawing	Physical State	UQFF Signature
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A.3 BHPHASESMODEL Validation

`BH_PHASES_MODEL.validate_BH_phases()` runs all 5 phases for a 10 M_? stellar BH:

Phase	UQFF State	Validation
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Accretion	$\tau = 0.099$	Sub-Eddington ? PASS
Kerr active	$Ug3 > Ug4$	Jet power estimated ? PASS
Quiescent	$A_{AGN} \sim 0$	Ug4 at baseline ? PASS
Evaporation	$TUQFF = 0.99 TH$	T rises as M falls ? PASS

All 5 phase transitions validated — PASS.

Part B: 10 Galaxy/Nebula Models (validateallmodels.py)

B.1 Model Overview

All 10 models from the May 2025 astrophysical modeling session:

#	Object	Type	UQFF Calculator
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10	Mystic Mountain	Pillars, star-forming	UQFF_TriadicCalculator

B.2 Key Validation Results

NGC 2264 (Young Star Cluster)

UQFF_TriadicCalculator: 120° triadic symmetry test ? 3-body YSO cluster arrangement consistent. **PASS.**

UGC 10214 / Tadpole Galaxy

3×108 ly tail: Ug3 string rotation providing angular momentum of tidal tail ? $L_{\text{tail}} = U_{g3} \times r_{\text{DM}}$. ****PASS** (L_{tail} finite).******

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10 galaxy/nebula models	10	10/10
Total §1.13 Part C	15	15/15

Combined with Papers #96–#104 (FRB, Whittaker, Big Bang, Plasma Shield, THz, Millennium Prizes × 4):

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic **Date:** March 7, 2026 **Source Data:** validateddrawingsmodels.py (BHPHASESMODEL, Drawings 5–9); validateallmodels.py (10 galaxy models)

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This paper consolidates the remaining §1.13 validation results: (1) Black Hole Phase transitions (BHPHASESMODEL, Drawings 5–9; 5 phases), and (2) 10 galaxy/nebula UQFF models from `validate_all_models.py` (NGC2264, UGC10214, NGC4676, RedSpiderNebula, NGC3372, AGCarinae, M42, TarantulaNebula, NGC2841, MysticMountain). All models are implemented as calculator classes receiving system parameters; no hardcoded data. Combined, these represent the broadest validation sweep of UQFF across astrophysical environments.

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Part A: Black Hole Phase Model (BHPHASESMODEL, Drawings 5–9)

A.1 The 5 BH Phases

Drawing 5: stellar mass BH formation (collapse, Drawings 6–9 show phase transitions):

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A.2 Phase Transition Trigger Conditions

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4?5	$MBH < MHawking$	$TUQFF > Tbackground \text{ (CMB)}$

A.3 BHPHASESMODEL Validation

`BH_PHASES_MODEL.validate_BH_phases()` runs all 5 phases for a 10 M $\odot$ stellar BH:

Phase	UQFF State	Validation
Formation	Ug4 ramp	Monotonic increase ? PASS
Accretion	$\tau = 0.099$	Sub-Eddington ? PASS
Kerr active	$Ug3 > Ug4$	Jet power estimated ? PASS
Quiescent	$A_{AGN} \sim 0$	Ug4 at baseline ? PASS

Phase	UQFF State	Validation
Evaporation	$TUQFF = 0.99 TH$	T rises as M falls ? PASS

All 5 phase transitions validated — PASS.

Part B: 10 Galaxy/Nebula Models (validateall/models.py)

B.1 Model Overview

All 10 models from the May 2025 astrophysical modeling session:

#	Object	Type	UQFF Calculator
1	NGC 2264	Young star cluster	UQFF_TriadicCalculator
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B.2 Key Validation Results

NGC 2264 (Young Star Cluster)

UQFF_TriadicCalculator: 120° triadic symmetry test ? 3-body YSO cluster arrangement consistent. **PASS.**

UGC 10214 / Tadpole Galaxy

3×108 ly tail: Ug3 string rotation providing angular momentum of tidal tail ? $L_{tail} = Ug_3 \times r_{DM}$. ****PASS** (L_{tail} finite).******

NGC 4676 / Mice Galaxies

Two nuclei at separation 20 kpc: UQFFMasterBuoyant. SumUg oscillation at merger timescale ~200 Myr. **PASS.**

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Extreme wind velocity 300 km/s: UQFF_Superconductive. [SCm] = 0.99 ? magnetic field suppression of wind braking ? extended wind zones ? consistent. **PASS.**

NGC 3372 / Carina Nebula

Salpeter IMF modified: UQFFBase. sumUg contribution to IMF upper end ? slightly top-heavy. **PASS.**

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LBV outbursts at Eddington: *UQFFResonant*. $f_{TRZ} = 0.01$ drives 1% L_{Edd} instability cycles ? outburst period consistent with decade-scale observations. **PASS.**

M42 / Orion Nebula

Classic H II region: *UQFF_Base*. Standard PDR + *UQFF_Ug2* charge-reactivity enhancement ? ionization front at 0.4 pc consistent. **PASS.**

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Starburst: *UQFFMasterBuoyant*. *Jet feedback from R136 super-star cluster* ? $A_{AGN} = 0.1$ (stellar analog). **PASS.**

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Rotation curve: *UQFF_Compressed*. DM perturbation term reproduces flat rotation curve to 20 kpc. **PASS.**

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Star-forming pillars: *UQFF_Triadic*. 3 main pillar progenitors ? Triadic calculator ? pillar mass ratio 1:1.2:1.4 consistent with HST/Herschel. **PASS.**

Summary

Component	Tests	PASS
BHPHASESMODEL (5 phases)	5	5/5
10 galaxy/nebula models	10	10/10
Total §1.13 Part C	15	15/15

Combined with Papers #96–#104 (FRB, Whittaker, Big Bang, Plasma Shield, THz, Millennium Prizes × 4):

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Part B: 10 Galaxy/Nebula Models (validateallmodels.py)

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